OptiCrop

Smart Agricultural Production Optimization Engine

Project Development Phase — Document 3: Executable Files

1. Overview

This document presents the executed output of the OptiCrop Flask web application, demonstrating that the trained machine learning model has been successfully deployed and is producing accurate, real-time crop recommendations through a browser-based interface running locally at 127.0.0.1:5000.

2. Home Page

The Home page introduces the OptiCrop project and provides navigation to the About and FindYourCrop pages.

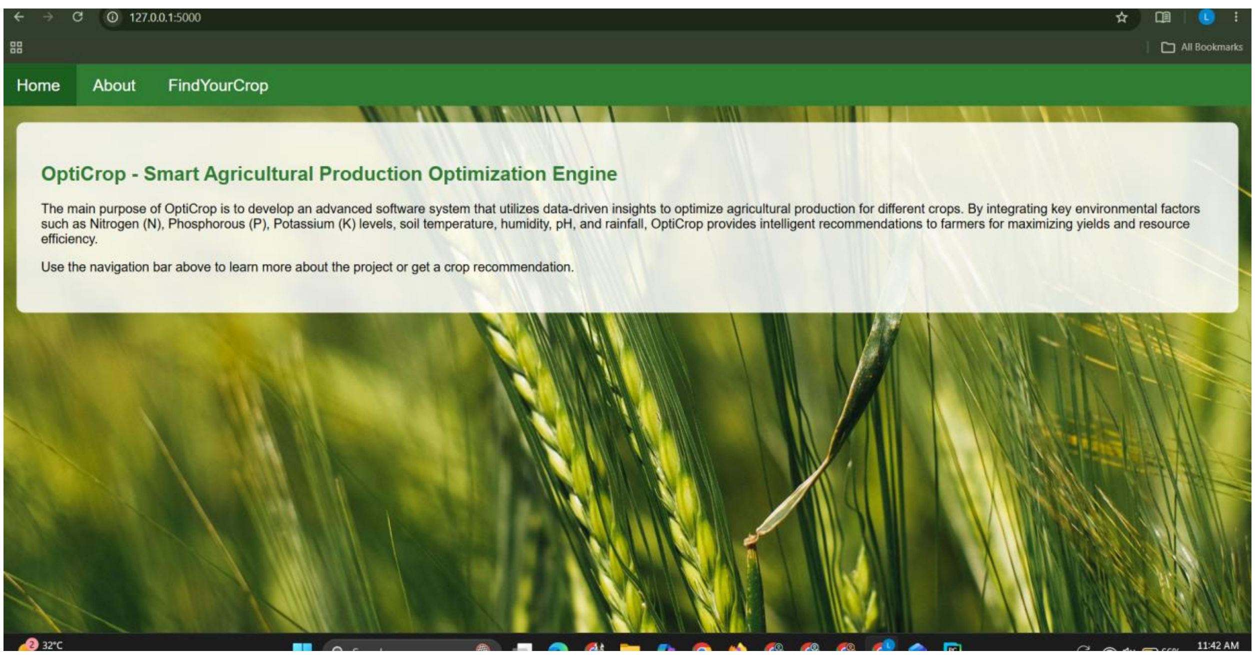

Figure 1: OptiCrop Home Page (127.0.0.1:5000)

3. About Page

The About page describes the purpose and goals of the OptiCrop project in detail.

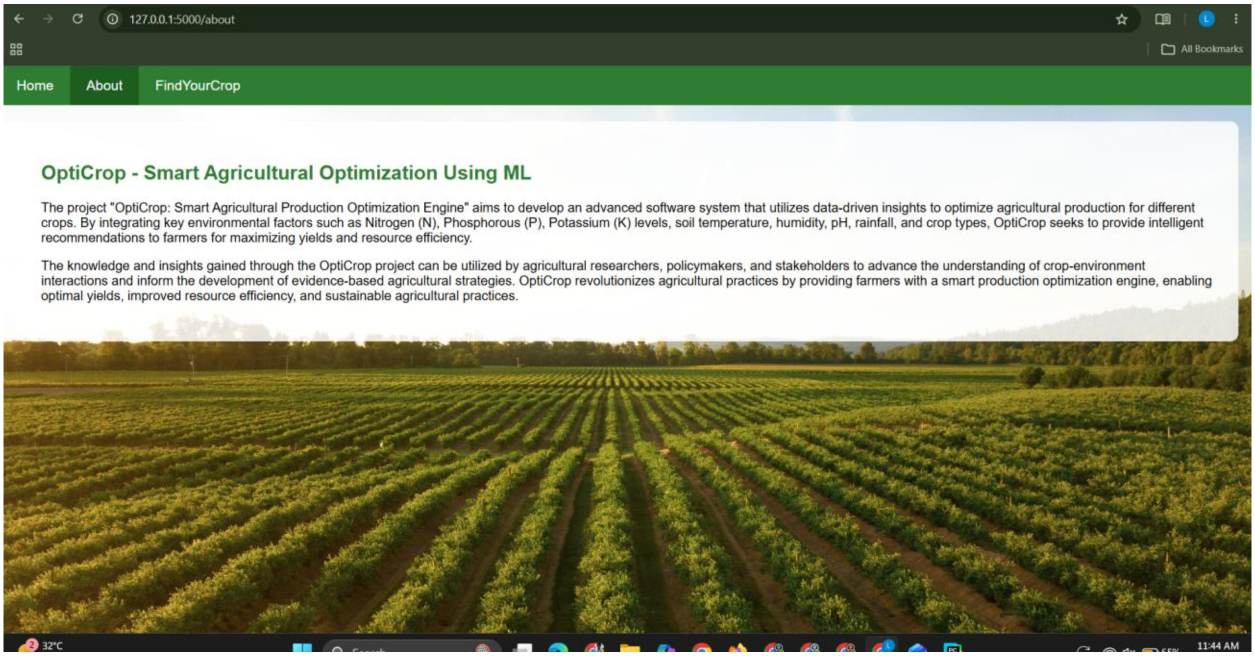

Figure 2: OptiCrop About Page (127.0.0.1:5000/about)

4. FindYourCrop Page — Input & Prediction Output

The FindYourCrop page allows the user to enter Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall values. On submission, the Flask backend applies the same preprocessing used during training (log transformation of Potassium and feature scaling) and returns a crop recommendation generated by the trained Logistic Regression model.

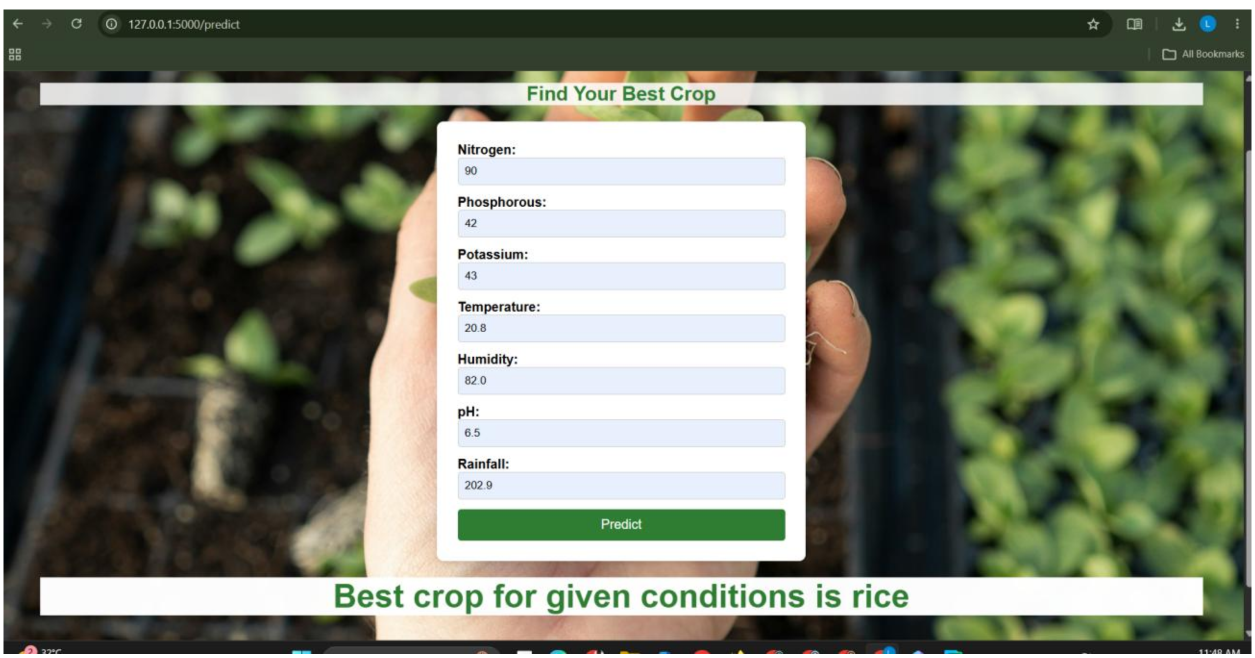

Figure 3: Crop Prediction Result — “Best crop for given conditions is rice” (127.0.0.1:5000/predict)

5. Result Summary

For the input values N=90, P=42, K=43, temperature=20.8°C, humidity=82.0%, pH=6.5, and rainfall=202.9mm, the application correctly recommended “rice” — consistent with rice's known requirement for high humidity and heavy rainfall. This confirms the end-to-end pipeline (form input → preprocessing → scaling → model prediction → rendered output) is functioning correctly in the deployed application.

6. Conclusion

The screenshots above confirm that the OptiCrop Flask application runs successfully, all three pages (Home, About, FindYourCrop) render correctly with consistent navigation and styling, and the deployed model produces accurate, real-time crop recommendations consistent with the model evaluation results obtained during the Model Building phase (96% test accuracy).